P-EDCA under light load: best P99 parameters vs saturated load

802.11be, nSta=30, CWds×QSRC×PSRC sweep | offered load 0.1 / 0.5 / 1.0 Mbps per STA | P-EDCA STA P99

The three load regimes (EDCA-only VO P99 sets the scene)

- **0.1 Mbps — uncongested**
  EDCA-only P99 = 0.92 ms (sub-ms). Nothing for P-EDCA to fix.
- **0.5 Mbps — congested, not saturated**
  EDCA-only P99 = 21.96 ms. Real queueing but the priority path can drain it.
- **1.0 Mbps — saturated**
  EDCA-only P99 = 46.23 ms. Collision-feedback regime.

0.1 Mbps: ≈ no-op

P-EDCA cuts P99 only +5–7%

best 0.87 ms vs EDCA 0.92 ms — any combo works (q0 even slightly worst)

0.5 Mbps: sweet spot

+87% P99 (22.0 → 2.9 ms)

QSRC=0, PSRC=3 — monotonic; worst combo ≤ 28 ms

One recipe at every load

QSRC = 0, PSRC = 3, CWds = 0/1

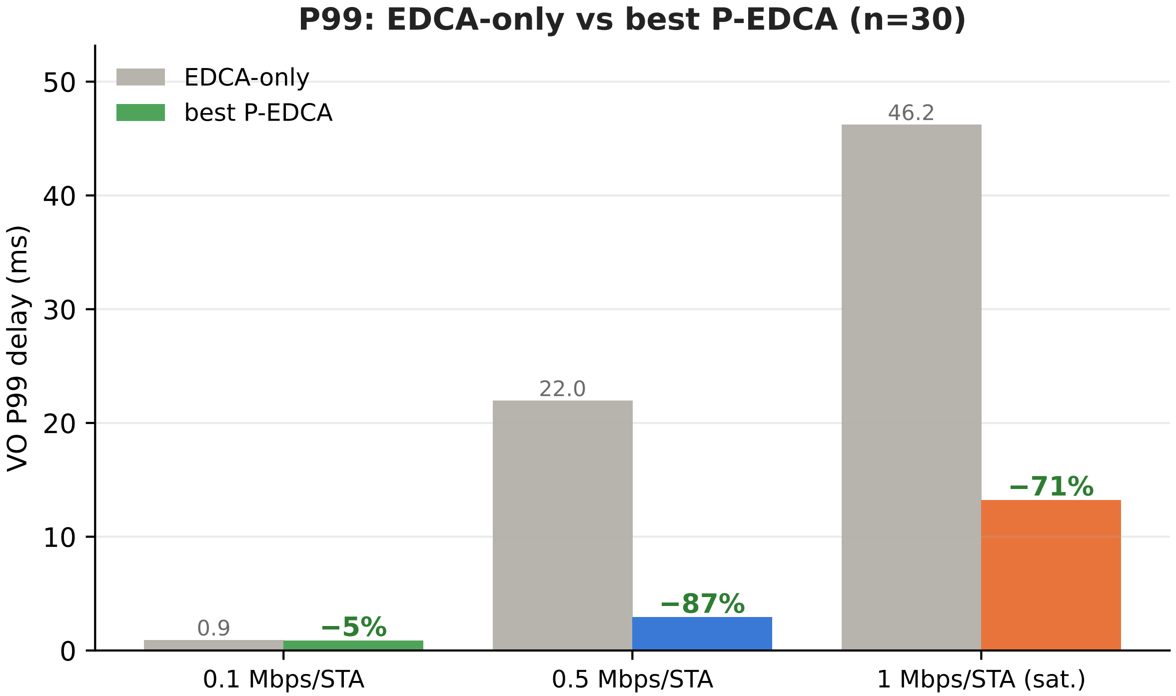
now wins at saturation too — the old n=5 tail blow-up is gone (v6.3.x fixes)

One-line takeaway

P-EDCA's value tracks how congested EDCA already is: near-zero at 0.1 Mbps, +87% at 0.5 Mbps and +71% at saturation (n=30). The aggressive recipe (QSRC 0, PSRC 3) wins at every load with real queueing — no tail risk left.

Best P-EDCA P99 across loads — the gain peaks at moderate load

EDCA-only baseline vs best-of-36-combos P-EDCA (P-EDCA STAs, n=30), with % P99 reduction



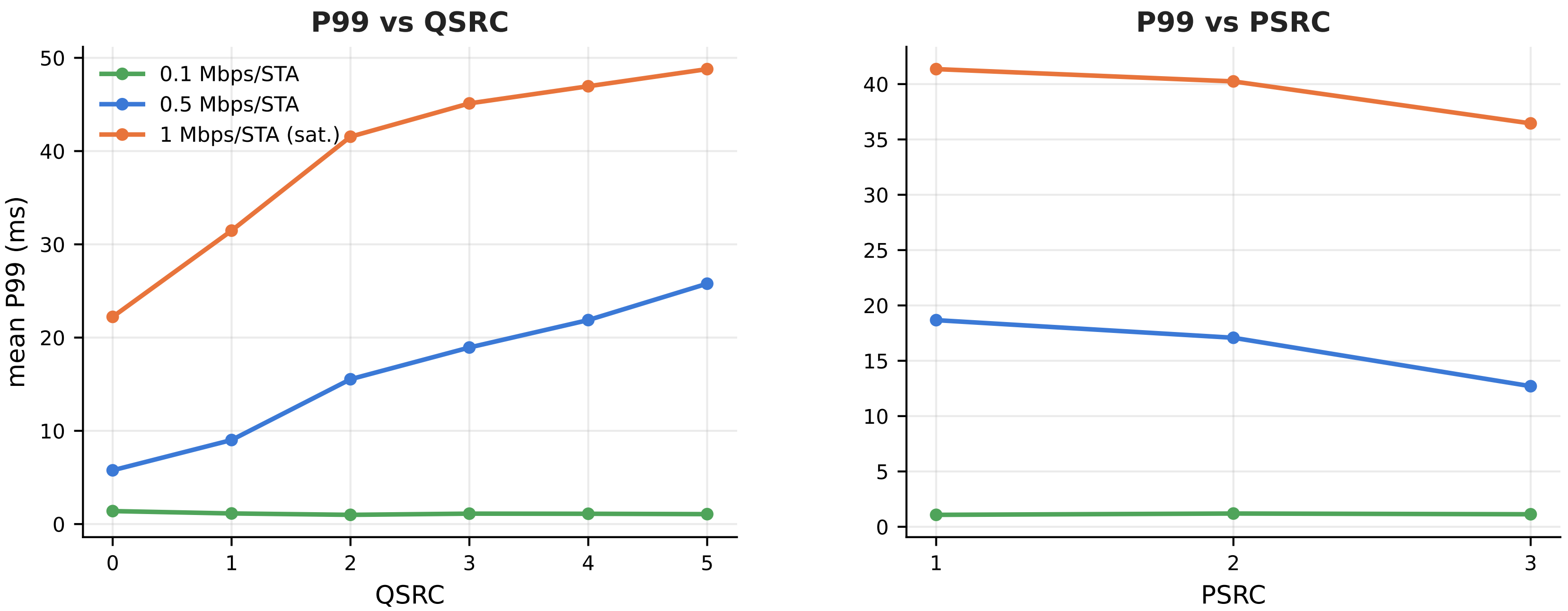
Best combo & P99 per P-EDCA population

Load	nPedca	best c/q/s	P99 (ms)
0.1 Mbps/STA	5	c0 q5 s2	0.86 (−7%)
	15	c1 q3 s1	0.86 (−7%)
	30	c0 q2 s3	0.87 (−5%)
0.5 Mbps/STA	5	c0 q1 s3	9.70 (−56%)
	15	c0 q0 s3	5.55 (−75%)
	30	c0 q0 s3	2.91 (−87%)
1 Mbps/STA (sat.)	5	c1 q0 s3	23.32 (−50%)
	15	c1 q0 s3	25.95 (−44%)
	30	c1 q0 s3	13.22 (−71%)

- Relative gain grows with congestion: +5–7% at 0.1 Mbps → +87% at 0.5 Mbps → +71% at saturation (n=30). P-EDCA only helps once EDCA itself is congested.
- At 0.1 Mbps the best P99 (0.87 ms) barely beats EDCA (0.92 ms) and the winning combo drifts across nPedca (c0 q5 s2 / c1 q3 s1 / c0 q2 s3) — a sign params don't matter when uncongested.
- At 0.5 & 1.0 Mbps the winner is consistently aggressive (QSRC 0–1, PSRC 3), and the gain RISES with penetration: 0.5 Mbps +56% → +75% → +87%; saturation +50% / +44% / +71% (n=5/15/30).

Which knob matters at which load — and where the old tail-risk went

Marginal mean P99 (P-EDCA STAs, n=30) vs QSRC and vs PSRC, per load



Tail blow-up: gone

worst-case P99 over 36 combos

0.1: 1.4 ms   0.5: 28 ms   1.0: 53 ms

worst ≈ EDCA-only baseline
(= no benefit, never a blow-up)

the pre-v6.3 294 ms n=5 blow-up no longer reproduces after the NAV/FEM fixes

- QSRC: at 0.1 Mbps a shallow reverse-U (q0 is mildly the WORST, 1.4 vs 1.0 ms at q2 — don't trigger P-EDCA when uncongested); at 0.5 Mbps strongly monotonic (q0 5.8 ms vs q5 25.8 ms); at 1.0 Mbps now ALSO monotonic (q0 22.2 ms vs q5 48.8 ms).
- PSRC: irrelevant at 0.1 Mbps; clearly larger-is-better at 0.5 Mbps (s3 12.7 ms vs s1 18.7 ms) and at saturation (s3 36.4 ms vs s1 41.4 ms) — with the v6.3.x fixes PSRC=3 no longer carries any blow-up risk.
- Robust recommendation — one recipe at every load with real queueing: QSRC=0, PSRC=3, CWds=0/1 (worst-case ≤ 28 ms at 0.5 Mbps, ≤ 53 ms at saturation ≈ the EDCA-only baseline). At 0.1 Mbps simply leave P-EDCA untriggered (large QSRC) — there is nothing to gain.